Unit 3.01: Diophantine Equations, SFFT, Bezout & Floor Functions

Q1. The coordinate diagram shows the parabolic curve f_1x = x^2 - 10x + 38 in the domain of Diophantine Equations, SFFT & Floor Functions. By completing the square f_1x = x - h^2 + k , find the sum k + f_17 .

[ ] A: 35
[ ] B: 17
[x] C: 30
[ ] D: 28

Solution: **Mark Scheme & Examiner Solution Diophantine Equations, SFFT & Floor Functions:** 1. Complete the square: x^2 - 10x + 38 = x - 5^2 - 25 + 38 = x - 5^2 + 13 , giving minimum value k = 13 . **[M1]** 2. Evaluate at x = 7 : f_17 = 2^2 + 13 = 17 . **[M1]** 3. Sum the two quantities: k + f_17 = 13 + 17 = 30 . **[A1]** - **Final Verified Answer:** 30 .

Q2. A mathematical model in Diophantine Equations, SFFT & Floor Functions is defined by the cubic function y_2x = 4x^3 - 5x^2 + 15x - 10 . Calculate the exact gradient of the tangent to the curve at the point where x = 3 .

[ ] A: 123
[ ] B: 78
[ ] C: 41
[x] D: 93

Solution: **Mark Scheme & Examiner Solution Diophantine Equations, SFFT & Floor Functions:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 12x^2 - 10x + 15 . **[M1]** 2. Substitute x = 3 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=3} = 129 - 103 + 15 = 93 . **[M1][A1]** - **Final Verified Answer:** 93 .

Q3. In Diophantine Equations, SFFT & Floor Functions, two displacement vectors are given by mathbf{u}_{3} = langle 5, -8 rangle and mathbf{v}_{3} = langle 9, 7 rangle . Compute the exact scalar value of mathbf{u}_{3} cdot mathbf{v}_{3} + |mathbf{u}_{3}|^2 .

[x] A: 78
[ ] B: -11
[ ] C: 89
[ ] D: 190

Solution: **Mark Scheme & Examiner Solution Diophantine Equations, SFFT & Floor Functions:** 1. Compute the inner product: mathbf{u}_{3} cdot mathbf{v}_{3} = 59 + -87 = 45 + -56 = -11 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{3}|^2 = 5^2 + -8^2 = 89 . **[M1]** 3. Combine: -11 + 89 = 78 . **[A1]** - **Final Verified Answer:** 78 .

Q4. The linear growth profile shown in the diagram for Diophantine Equations, SFFT & Floor Functions follows an arithmetic progression with first term a_1 = 6 and common difference d = 6 . Find the exact sum S_{15} of the first 15 terms.

[ ] A: 90
[x] B: 720
[ ] C: 810
[ ] D: 1350

Solution: **Mark Scheme & Examiner Solution Diophantine Equations, SFFT & Floor Functions:** 1. Use the sum formula for an arithmetic series: S_n = frac{n}{2}[2a_1 + n - 1d] . **[M1]** 2. Substitute n = 15, a_1 = 6, d = 6 : S_{15} = frac{15}{2}[12 + 14 times 6] = frac{15}{2} times 96 = 720 . **[M1][A1]** - **Final Verified Answer:** 720 .

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Q5. In Diophantine Equations, SFFT & Floor Functions, evaluate the exact definite integral I_5 = int_0^{2} 18x^2 + 20x,mathrm{d}x .

[ ] A: 112
[ ] B: 108
[x] C: 88
[ ] D: 45

Solution: **Mark Scheme & Examiner Solution Diophantine Equations, SFFT & Floor Functions:** 1. Integrate term by term: int 18x^2 + 20x,mathrm{d}x = left[6x^3 + 10x^2right]_0^{2} . **[M1]** 2. Substitute the upper and lower limits: 62^3 + 102^2 - 0 = 48 + 40 = 88 . **[M1][A1]** - **Final Verified Answer:** 88 .

Q6. In Diophantine Equations, SFFT & Floor Functions, consider the quadratic function f_6x = x^2 - 12x + 121 . Express f_6x in the completed-square form x - h^2 + k and compute the exact value of k + f_613 .

[ ] A: 225
[ ] B: 134
[ ] C: 212
[x] D: 219

Solution: **Mark Scheme & Examiner Solution Diophantine Equations, SFFT & Floor Functions:** 1. Complete the square: x^2 - 12x + 121 = x - 6^2 - 36 + 121 = x - 6^2 + 85 , giving minimum value k = 85 . **[M1]** 2. Evaluate at x = 13 : f_613 = 7^2 + 85 = 134 . **[M1]** 3. Sum the two quantities: k + f_613 = 85 + 134 = 219 . **[A1]** - **Final Verified Answer:** 219 .

Q7. The graph illustrates the cubic model y_7x = 4x^3 - 6x^2 + 33x - 17 studied in Diophantine Equations, SFFT & Floor Functions. Find the instantaneous gradient frac{mathrm{d}y}{mathrm{d}x} of the tangent to the curve at x = 8 .

[x] A: 705
[ ] B: 801
[ ] C: 672
[ ] D: 117

Solution: **Mark Scheme & Examiner Solution Diophantine Equations, SFFT & Floor Functions:** 1. Differentiate term by term using the power rule: frac{mathrm{d}y}{mathrm{d}x} = 12x^2 - 12x + 33 . **[M1]** 2. Substitute x = 8 : left.frac{mathrm{d}y}{mathrm{d}x}right|_{x=8} = 1264 - 128 + 33 = 705 . **[M1][A1]** - **Final Verified Answer:** 705 .

Q8. In Diophantine Equations, SFFT & Floor Functions, two displacement vectors are given by mathbf{u}_{8} = langle 6, -7 rangle and mathbf{v}_{8} = langle 8, 5 rangle . Compute the exact scalar value of mathbf{u}_{8} cdot mathbf{v}_{8} + |mathbf{u}_{8}|^2 .

[ ] A: 13
[x] B: 98
[ ] C: 85
[ ] D: 168

Solution: **Mark Scheme & Examiner Solution Diophantine Equations, SFFT & Floor Functions:** 1. Compute the inner product: mathbf{u}_{8} cdot mathbf{v}_{8} = 68 + -75 = 48 + -35 = 13 . **[M1]** 2. Compute the squared norm: |mathbf{u}_{8}|^2 = 6^2 + -7^2 = 85 . **[M1]** 3. Combine: 13 + 85 = 98 . **[A1]** - **Final Verified Answer:** 98 .